

JavaScript Questions-2

Arrow Functions

Instructions

Convert each of the following normal functions into Arrow Functions.

1. Find the Cube

```
function cube(num) {  
    return num * num * num;  
}
```

2. Check Eligibility

```
function isEligible(age) {  
    return age >= 18;  
}
```

3. Generate Welcome Message

```
function welcome(name) {  
    return `Welcome ${name}`;  
}
```

4. Find the Larger Number

```
function getMax(a, b) {  
    return a > b ? a : b;  
}
```

5. Calculate Rectangle Area

```
function area(length, width) {  
    return length * width;  
}
```

6. Print a Message

```
function showMessage() {  
    console.log("Learning JavaScript");  
}
```

7. Create a Product Object

```
function createProduct(name, price) {  
    return {  
        name: name,  
        price: price  
    };  
}
```

Callbacks

Instructions

Complete the following programs using callbacks.

1. Greeting Callback

Create a function `welcomeUser()` that accepts a callback and passes the name "Rahul" to it.

Expected Output:

Hello Rahul

2. Number Processor

Create a function `processNumber()` that sends the value 25 to a callback function.

Expected Output:

Number received: 25

3. Order Confirmation

Create a function `placeOrder()` that prints:

Order Placed

Then executes a callback that prints:

Order Confirmed

4. Student Result

Create a function `checkResult()` that passes "Pass" to a callback.

Expected Output:

Result: Pass

5. Movie Ticket Booking

Create a function bookTicket() that prints:

Booking Ticket...

Then executes a callback that prints:

Ticket Booked Successfully

6. Favorite Color

Create a function getColor() that passes "Blue" to a callback function.

Expected Output:

My favorite color is Blue

7. Delayed Message

Using setTimeout() and a callback, display:

Loading...

and after 2 seconds display:

Welcome Back!

Callback Methods

Instructions

Solve the following problems using Array Iteration Callback Methods. You may use one or more callback methods wherever appropriate.

1. Online Shopping Cart

An e-commerce website stores product prices in an array.

```
const prices = [499, 999, 1499, 299, 799];
```

Create a new array containing the prices after applying a 10% discount on every product.

Expected Output:

```
[449.1, 899.1, 1349.1, 269.1, 719.1]
```

2. Student Pass List

A coaching institute stores marks of students in an array.

```
const marks = [35, 78, 90, 25, 60, 48, 82];
```

Create a new array containing only the students who have scored 50 or above.

Expected Output:

```
[78, 90, 60, 82]
```

3. Employee Salary Report

A company stores employee salaries in an array.

```
const salaries = [25000, 30000, 45000, 50000, 35000];
```

Find the total amount the company spends on salaries every month.

Expected Output:

185000

4. Product Search

An online store maintains a list of product names.

```
const products = [
```

```
  "Laptop",
```

```
  "Mobile",
```

```
  "Headphones",
```

```
  "Keyboard",
```

```
  "Mouse"
```

```
];
```

A customer searches for "Keyboard". Find and display the matching product.

Expected Output:

Keyboard

5. Movie Ratings Dashboard

A movie website stores ratings in an array.

```
const ratings = [4.2, 3.8, 4.9, 2.5,  
4.5, 3.2];
```

Arrange the ratings from highest to lowest.

Expected Output:

```
[4.9, 4.5, 4.2, 3.8, 3.2, 2.5]
```

6. User Profile Cards

A website stores user names in an array.

```
const users = ["Rahul", "Priya",  
"Kiran", "Sneha"];
```

Create a new array where each name is converted into the following format:

Expected Output:

```
[  
  "Welcome Rahul",  
  "Welcome Priya",  
  "Welcome Kiran",  
  "Welcome Sneha"  
]
```

7. Food Delivery App

A food delivery application stores orders.

```
const orders = [  
  { id: 1, status: "Delivered" },  
  { id: 2, status: "Pending" },  
  { id: 3, status: "Delivered" },  
  { id: 4, status: "Cancelled" },  
  { id: 5, status: "Pending" }  
];
```

Create a new array containing only the orders whose status is "Pending".

Expected Output:

```
[  
  { id: 2, status: "Pending" },  
  { id: 5, status: "Pending" }  
]
```

Bonus Challenge

Company Employee Analytics

```
const employees = [  
  { name: "Rahul", salary: 30000 },  
  { name: "Priya", salary: 45000 },  
  { name: "Kiran", salary: 25000 },  
  { name: "Sneha", salary: 50000 }  
];
```

Perform the following tasks:

1. Create a new array containing only employee names.
2. Find the employee earning ₹50,000.
3. Calculate the total salary of all employees.
4. Arrange employees in ascending order of salary.